

Load Frequency Control for Integrated Electrical-Water System Considering Water Hammer Effect

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Abstract—Demand response (DR) is a valuable tool in load frequency control (LFC) of power systems, especially in improving the system’s regulating capacity. The pumps, with their rapid response capabilities, can be effectively utilized as a DR resource to further enhance LFC performance. However, improper adjustment of pumps will cause the accumulation of pressure pulsations throughout the water distribution system (WDS), resulting in the water hammer effect. To address this issue, this paper introduces a water hammer feedback loop in LFC with DR to balance the power system stability as well as WDS operational safety. With the wave tracking method (WTM), the proposed LFC method enables efficient modeling of the hydrodynamic process with low computational complexity and acceptable accuracy. A model predictive control (MPC) based optimal control algorithm is also developed to adjust the pumps and generation units. Case studies demonstrate that the proposed model exhibits significant improvement on the performance of pressure pulsation regulation. Results also show the enhanced stability and effective pressure fluctuation mitigation in the electric-water coordinated control.

Index Terms—Electric-water coordinated control, demand response, water hammer effect, load frequency control, model predictive control.

I. INTRODUCTION

LOAD frequency control (LFC) maintains the power system stability by matching the supply and demand in response to variations in customer demands and volatile renewable energy. The accelerating transition into clean energy continually reduces the proportion of thermal generating units [1]. It exacerbates the shortage of system regulating

capacity and reserve resources. Consequently, there is a pressing need to investigate the resources the demand side can contribute to system stability.

Water distribution system (WDS), as an energy-intensive infrastructure, holds significant potential for flexible regulation. WDS accounts for 7.1% of the total annual energy consumption in California, USA [2] and up to 10% in Changzhou City, China [3]. Frequency support capability of the WDS is mainly derived from the rapid response characteristics and inertial support of water pumps. The adjustment range is continuously expended with the increasing adoption of variable-speed pumps (VSPs) [4]. Since VSPs are connected to the grid via the frequency inverters, their contribution to inertial support is reduced or non-existent [5]. This means their operation must be explicitly controlled, cannot autonomously react to frequency fluctuation. Their response is orchestrated by the DR aggregator based on the control signal from the control facility of power systems [6].

Pump control is a key step to stability operation of WDS. WDS exhibits “high inertia” and is slow to respond to disturbances or adjustments, owing to the large mass of water within it. “High inertia” increases system stability by buffering against sudden changes. Thus, frequency fluctuation of power system can be partly transferred within the WDS by DR control of pumps. However, Changes in pump speed bring pressure pulsations, known as the water hammer effect, which propagates as hydroacoustic pressure waves throughout the WDS [18]. The overly rapid response of VSPs causes drastic pressure fluctuations in WDS, leading to the vibration issues that may jeopardize the system’s safety [19]. Existing studies treat water hammer as an isolated issue. Those works focus on pipe design optimization [29], active noise control [20], and pressure relief valves [32]. The strategies primarily aim to accelerate the attenuation of water hammer by increasing the damping in the water transmission process. The dynamic interactions between water hammer mitigation and the role of pumps in grid stability were not considered. In fact, the water hammer effect can be reduced at its source through proper pump control. It is essential to balance the stability performance of the power system and water distribution system when more water pumps are incorporated into LFC as DR resources.

The ancillary frequency services provided by DR have been developed [7], [8], [9]. The flexible regulation of DR increases

Received 23 December 2024; revised 24 April 2025; accepted 26 May 2025. Date of publication 3 June 2025; date of current version 25 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 52177073, and in part by the Fundamental Research Funds for the Central Universities under Grant 2025CDJZKYYJH-15 and Grant 2023CDJYJH045. Paper no. TSG-02256-2024. (*Corresponding author: Chunyan Li.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSG.2025.3576195>.

Digital Object Identifier 10.1109/TSG.2025.3576195

the stability margin and minimizes the frequency fluctuation of power system [10], [11]. Many LFC control methods for DR were also proposed, such as PID control [12], adaptive control [13], model predictive control [14], and sliding mode control [15]. Control strategy for pump participating in the frequency control was proposed to provide the frequency regulation services [16], [17]. Most works on LFC with pumps assume that the safety of WDS is inherently satisfied with the limited output range without considering the water hammer effect of WDS.

To the best of our knowledge, the modeling of water hammer effect and its use in the LFC with pumps have not been reported.

The hydrodynamic process of water hammer effect can be described by the differential momentum and the continuity equations, which are generally the partial differential equations (PDEs) that do not have analytic solutions [21]. Numerical methods, such as the finite difference (FD) method [22] and the method of characteristics (MOC) [23], have been used to find the solutions. These methods are often applied in the form of time and space discretization, described as "characteristic grids". Under actual operating conditions, dense spatial discretization, which consumes substantial computational resources, may not be necessary. Operators typically focus on specific issues, such as local pressure surges at pumps or other critical points [24]. Recently, the electric circuit analogy has been applied to model the water transient process [25]. But that study relies on Laplace transform, which makes the model parameterization relatively challenging.

Most studies for LFC with DRs were developed based on the idea that the consumption of controllable loads can be changed instantaneously to improve the frequency stability of power system [6]. In these works, DRs were treated as independent units with constant frequency response dynamics. This is not the case for the pumps in WDS, because the control of one pump affects the water flow and pressure of other pumps. Thus, pumps need to be considered as interconnected units in the WDS model to better reflect the dynamic response of pump control. Fast pump control enables fast response for frequency stability of the power system. However, it also causes abrupt change in water pressure, which leads to the water hammer effect. In a situation that requires the control of multiple pumps, improper control can incur the cumulation of water pressure fluctuations, resulting in serious hammer effects.

In this paper, a coordinated LFC model for the integrated electrical-water system (IEWS) is proposed to enhance LFC performance by achieving fast frequency stability of power system and, at the same time, alleviating the water pressure fluctuation. The model considers pump pressure establishes a water hammer feedback loop is developed to deal with the above problem and to enhance the LFC performance. The wave tracking method (WTM) [26] is employed to simulate the transient process and the water hammer effect in WDS with low computational complexity and high accuracy. Main contributions are

a) Propose an electric-water coordinated load frequency control framework considering the water hammer effect in

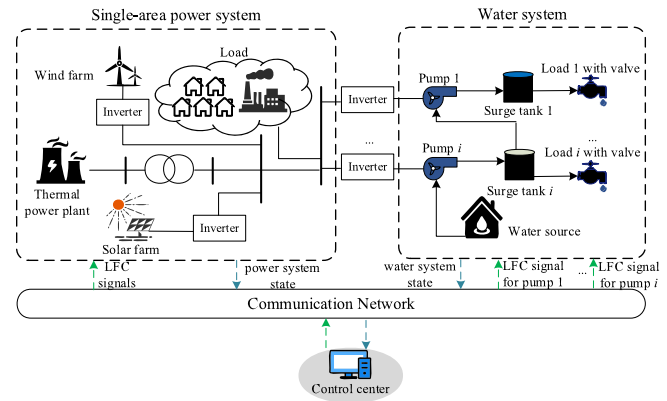


Fig. 1. Schematic diagram of a single-area power system with the water system.

WDS. This framework introduces a water hammer feedback loop, fully leveraging the "high inertia" of the WDS.

b) Establish a recursive equation based on WTM to describe the dynamic response of pump pressure and derive the steady-state solution. It provides a realistic representation of the water hammer effect.

c) Develop an MPC based controller to balance the performance of frequency stability in the power system and the pressure safety in WDS, and improve the frequency response and alleviate the water hammer effect.

The remaining sections are as follows. Section II proposes a coordinated control model of IEWS. Section III formulates the controller design of IEWS-LFC model, derives the analytic solution using the MPC approach, and analyzes the control stability over an infinite horizon. Section IV presents the case study results to validate the efficiency of WTM and benefits of the proposed MPC control on stability performance, followed by Conclusion in Section V.

II. PROBLEM FORMULATION FOR INTEGRATED ELECTRICAL-WATER SYSTEM

This section describes how the water distribution system (WDS) plays in the load frequency control (LFC) of power systems to improve the stability. First, the dynamic response model for WDS is constructed based on the wave tracking method. Then an IEWS-LFC model with water hammer feedback loop is developed.

Fig. 1 depicts the electrical water system structure. It shows the connections between the single-area power system and water system in the integrated power and water distribution system (IEWS). It can be seen from Fig. 1 that the WDS is coupled with the power system via pumps.

The LFC model of an integrated electrical-water system (IEWS) with DRs is shown in Fig. 2. This model contains three frequency control loops: the primary control, supplementary control, and DR control loops. This study introduces a WDS and the water hammer feedback loop in the DR control. To accommodate for safety requirement of the WDS operation, water hammer feedback control is deployed to each pump. The sharing rate $1-\alpha$ represent the agreement of the WDS operators in frequency services [8]. The pump, used as DR resource

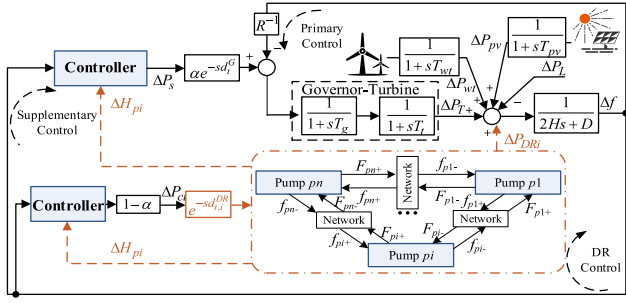


Fig. 2. An LFC model of the integrated electrical-water system with DRs.

ΔP_{DRi} , mimics the spinning reserve to suppress the frequency oscillations [2].

Pumps, when used as DR resources, provide continuous load regulation by adjusting the rotating speed. The primary and supplementary control loops can be derived from the simplified power system model [7]. Parameters T_g and T_t represent the equivalent governor and turbine time constants, respectively. R is the equivalent drop constant. T_{wt} and T_{pv} are the time constants for the wind turbine and photovoltaic system, respectively. The aggregate delays of control channels in the conventional generation and DR are denoted as dG t and dDR t, i , which can be estimated a priori [9]. When the frequency deviation is negative (positive), the pump's consumption decreases (or increases) by variable speed drives [4].

The LFC model of IEWS can be modified as the power balance equation, including pumps [12]:

$$\Delta P_T(s) + \Delta P_{wt}(s) + \Delta P_{pv}(s) - \Delta P_L(s) + \sum_i^{N_p} \Delta P_{DRi}(s) = (2Hs + D)\Delta f(s) \quad (1)$$

In (1), $\Delta f(s)$, $\Delta P_T(s)$, $\Delta P_{wt}(s)$, $\Delta P_{pv}(s)$, $\Delta P_L(s)$, $2H$, D , and $\Delta P_{DRi}(s)$ are the frequency deviation, turbine output deviation, wind power output change, solar power output change, load change, inertia constant, and the equivalent load-damping coefficient, respectively.

Generally, there are no limitations on the ramp-up and ramp-down of the DR resources [10]. However, the water hammer effect, caused by the rotating speed regulation of pumps, impairs the safety of the WDS [18]. It propagates as hydroacoustic pressure waves (i.e., f_{pi} and F_{pi}) throughout the WDS network. The DR control of pumps is required to be constrained based on the magnitude of pressure pulsation ΔH_{pi} , which could affect the power system LFC performance. Thus, the primary challenge for IEWS control is simulating the influence of demand changes in pumps by modeling the dynamic response.

A. Dynamic Response Model for WDS

The dynamic response of WDS is modeled to represent the relationship between the pressure waves in the pipes and the water hammer effect on the node components during the pump adjustment process. The wave tracking method (WTM) for hydraulic systems is a powerful tool for modeling the hydraulic

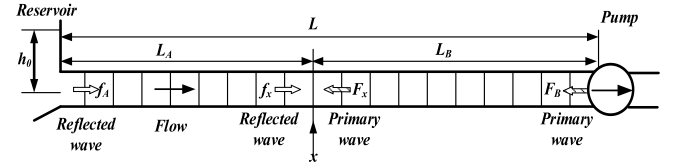


Fig. 3. Illustration of the wave tracking method.

dynamic response [27]. WTM can be used to analyze the water hammer effects in WDS.

Based on the fact that the flow velocity (c) is considerably smaller than the hydroacoustic pressure wave velocity (a), the wave equation can be derived from the hydraulic transient equations [26]

$$\frac{d}{dx} \left[h - h_0 \pm \frac{a}{gA} (Q - Q_0) \right] = -\frac{\lambda}{2gdA^2} (Q|Q| - Q_0|Q_0|). \quad (2)$$

h is the pressure head and Q the volume flow rate. The subscript 0 denotes the initial state, which can be obtained by the Newton-Raphson method. λ is the friction coefficient, g the gravitational constant, d the inner diameter of the pipe, and x the position along the pipe. A denotes the cross-section area of the pipe. Its value is fixed.

Suppose that the pipes are filled with water. The shock wave dynamics can be expressed as

$$2f(x - at) = h - h_0 + \frac{a}{gA} (Q - Q_0) \quad (3)$$

$$2F(x + at) = h - h_0 - \frac{a}{gA} (Q - Q_0). \quad (4)$$

$f(x - at)$ denotes a pressure shock wave propagating with velocity a along the water flow direction, while $F(x + at)$ denotes the pressure shock wave propagating in the opposite direction.

Fig. 3 illustrates the operation of the wave tracking method, which considers the interplay of different pressure shock waves. In WDS, dynamics between the primary wave and the reflected wave are quantitatively analyzed. In particular, the flow change at point x in the pipe is explored by adjusting the pump at the end of the pipe. It can be seen from Fig. 3 and (4) that the change of the flow rate close to the pump introduces the primary wave, denoted as F_B . This primary wave propagates with a velocity a and travels a distance L to reach the reservoir. At the reservoir, the pressure remains constant at h_0 . Each arrived pressure wave will be balanced by a reflected wave f_A such that $f_A = -F_A$. The reflected wave then propagates back in the pipe.

As the reflected wave f_A travels through a section of length L_A , it will meet with a newly generated primary wave propagating through a section of length L_B at point x . The propagation process can be approximated by using the mean value theorem on (2) to obtain

$$F_x(t + d_B^w) = F_B(t) + \frac{1}{2} R_B (\bar{Q}(t) |\bar{Q}(t)| - Q_{x,0} |Q_{x,0}|) \quad (5)$$

$$f_x(t + d_A^w) = f_A(t) - \frac{1}{2} R_A (\bar{Q}(t) |\bar{Q}(t)| - Q_{x,0} |Q_{x,0}|), \quad (6)$$

where $h_{x,0}$ and $Q_{x,0}$ are the initial pressure and flow rate at point x , respectively. $\bar{Q}(t)$ is the average of volume flow

rate during $[t, t+d_{A(B)}^w]$. $R_{A(B)} = \lambda L_{A(B)} / 2gdA^2$ are the resistance constants for f_x (F_x) and $d_{A(B)}^w = L_{A(B)} / a$ are the delay factors for f_x (F_x). The pressure h_x and flow Q_x at point x can be determined by the superposition of the primary wave F_x and the reflected wave f_x

$$\begin{aligned} h_x(t) &= f_x(t) + F_x(t) + h_{x,0} \\ Q_x(t) &= gA(f_x(t) - F_x(t))/a + Q_{x,0}. \end{aligned}$$

Then the dynamic response model of WDS can be represented in terms of the shock waves described below.

$$\begin{aligned} \frac{\eta_{pi} P_{pi}(t)}{\rho g Q_{pi}(t)} - \frac{a(A_{pi,+} + A_{pi,-})}{gA_{pi,+}A_{pi,-}} Q_{pi}(t) - \Delta h_{pi,0}^* \\ = 2(F_{pi,-}(t) - f_{pi,+}(t)) \end{aligned} \quad (7a)$$

$$h_{pi,+(-)}(t) = F_{pi,+(-)}(t) + f_{pi,+(-)}(t) + h_{pi,0,+(-)} \quad (7b)$$

$$h_{vi}(t) + k_{vi}\sqrt{h_{vi}(t)} - (h_{vi,0}^* + 2f_{vi}(t)) = 0 \quad (7c)$$

$$h_{vi,0}^* = h_{vi,0} + \frac{a}{gA_{vi}} Q_{vi,0} \quad (7d)$$

$$F_{vi}(t) = -f_{vi}(t) + h_{vi}(t) - h_{vi,0}. \quad (7e)$$

$$\sum_i^{\mathbb{J}_+} (F_i(t) - f_i(t))A_i = \sum_j^{\mathbb{J}_-} (F_j(t) - f_j(t))A_j \quad (7f)$$

$$F_i(t) + f_i(t) = F_j(t) + f_j(t), \forall i, j \in \mathbb{J} \quad (7g)$$

$$h_{Ti}(t + \Delta t) - h_{Ti}(t) = (Q_{Ti,+}(t) - Q_{Ti,-}(t))\Delta t / A_{Ti} \quad (7h)$$

$$F_{Ti}(t) = -f_{Ti}(t) + h_{Ti}(t) - h_{Ti,0} \quad (7i)$$

$$Q_{Ti,+(-)}(t) = \frac{gA_{Ti,+(-)}}{a} (f_{Ti,+(-)}(t) - F_{Ti,+(-)}(t)) + Q_{Ti,0,+(-)} \quad (7j)$$

Equations (7a) describes the relationship between the shock waves, the injected power, and the flow in a pump. The term $\Delta h_{pi,0}^*$ can be obtained by

$$\Delta h_{pi,0}^* = h_{pi,0,-} - h_{pi,0,+} - aQ_{pi,0}(1/A_{pi,+} + 1/A_{pi,-})/g.$$

Equation (7b) expresses the relationship between the shock waves and the water head in a pump. The dynamic response of shock waves at the valve can be obtained by (7c)–(7e). Equations (7f) and (7g) represent the balance of flow and water head in the junction of pipes, respectively. The flow and head are replaced with the shock waves of (3) and (4). Equations from (7h) to (7j) model the surge tank.

The plus or minus sign in the subscript of (7a) denotes the inlet or outlet to the pump. ρ is the density of water. Suppose that the conversion efficiency η_{pi} is constant. Then, equation (7a) can be implied by the pump's electricity consumption equation [25], equation (3) and equation (4). The set of a junction $\mathbb{J} = (\mathbb{J}_+, \mathbb{J}_-)$ with $\mathbb{J}_+$ and $\mathbb{J}_-$ represents the set of the inlet waves and that of the outlet waves, respectively. The reflection process represented by (7i) occurs inside the surge tank. F_{Ti} and f_{Ti} are the primary and the reflected waves inside the tank, respectively. In (7j), the plus or minus symbol indicates the direction of flow into or out of the tank at the steady state. k_{vi} is the coefficient of valve, which is related to the valve type and opening [23].

This study employs the MPC method to alleviate the water hammer effect and balance the performance of WDS and power system. This method allows for a customized cost function, enabling the optimization of specific objectives, and requires a controller design based on future states of WDS

and power system. Unlike traditional control methods such as PI or rule-based control, MPC explicitly incorporates system dynamics, input/output constraints, and future reference trajectories into its optimization framework. This approach enables more precise and proactive control and can regulate the load frequency and water hammer more efficiently than other controllers. The future states of pressure pulsations in WDS are obtained by (7a)–(7j). Since equations (5), (6), (7a), and (7c) are nonlinear, the dynamics of water hammer effect can be linearized around an equilibrium point for stability analysis and controller design [6].

Expanding the Taylor approximation of $\bar{Q}(t)|\bar{Q}(t)|$ around the initial flow rate $Q_{x,0}$, we have

$$\bar{Q}(t)|\bar{Q}(t)| \approx 2|Q_{x,0}|\bar{Q}(t) - Q_{x,0}|Q_{x,0}|.$$

Substituting it into (5) and (6) yields

$$F_x(t + d_B^w) = F_B(t) + R_B|Q_{x,0}|(\bar{Q}(t) - Q_{x,0}) \quad (8)$$

$$f_x(t + d_A^w) = f_A(t) - R_A|Q_{x,0}|(\bar{Q}(t) - Q_{x,0}). \quad (9)$$

In (7a), suppose that $Q_{pi}(t) = Q_{pi,0} + \Delta Q_{pi}$, where the $Q_{pi,0}$ is the initial flow of the pump. Then,

$$1/Q_{pi}(t) = 1/(Q_{pi,0} + \Delta Q_{pi}) = (1/Q_{pi,0})/(1 + \Delta Q_{pi}/Q_{pi,0}).$$

For the small-signal model, $\Delta Q_{pi}/Q_{pi,0} \ll 1$. Thus, $1/(Q_{pi,0} + \Delta Q_{pi}) \approx 1/Q_{pi,0}$. The equation (7a) can be linearized to

$$\begin{aligned} \frac{\eta_{pi} P_{pi}(t)}{\rho g Q_{pi,0}} - \frac{a(A_{pi,+} + A_{pi,-})}{gA_{pi,+}A_{pi,-}} Q_{pi}(t) - \Delta h_{pi,0}^* \\ = 2(F_{pi,-}(t) - f_{pi,+}(t)) \end{aligned} \quad (10)$$

In (7h), Taylor approximation is expanded around the initial pressure $h_{vi,0}$ so that

$$k_{vi}\sqrt{h_{vi}(t)} \approx k_{vi}\sqrt{h_{vi,0}} + \frac{0.5k_{vi}(h_{vi}(t) - h_{vi,0})}{\sqrt{h_{vi,0}}}.$$

Substituting it into the equation (7h), we obtain

$$h_{vi}(t) + k_{vi}(\sqrt{h_{vi,0}} + \frac{h_{vi}(t) - h_{vi,0}}{2\sqrt{h_{vi,0}}}) - (h_{vi,0}^* + 2f_{vi}(t)) = 0. \quad (11)$$

The linear recursive equations from (7b), (7d)–(7j), and (8) to (11) can be converted into a compact form of the state-space representation:

$$\bar{A}_{w,0}x_w(t) + \sum_{m=1}^{N_m} \bar{A}_{w,m}x_w(t - d_m^w) + \bar{B}_w u_{DR}(t) = x_{w,0} \quad (12)$$

where $x_w(t) = [x_{w1}(t), x_{w2}(t), \dots, x_{wN}(t)]^T$ is the state vector of WDS, $x_{wi}(t) = [F_i, f_i, Q_{pi}, F_{pi}, f_{pi}, h_{pi}, F_{Ti}, f_{Ti}, h_{Ti}, Q_{Ti}, F_{vi}, f_{vi}, h_{vi}]^T$, $u_{DR}(t) = [u_{DR1}(t), u_{DR2}(t), \dots, u_{DRN}(t)]^T$ is the control input of the pumps, and $x_{w,0}$ is the constant vector caused by initial state. d_w is the propagating delay of shock wave in pipe m . $\bar{A}_w$ is the state transition matrix and $\bar{B}_w$ the input matrix.

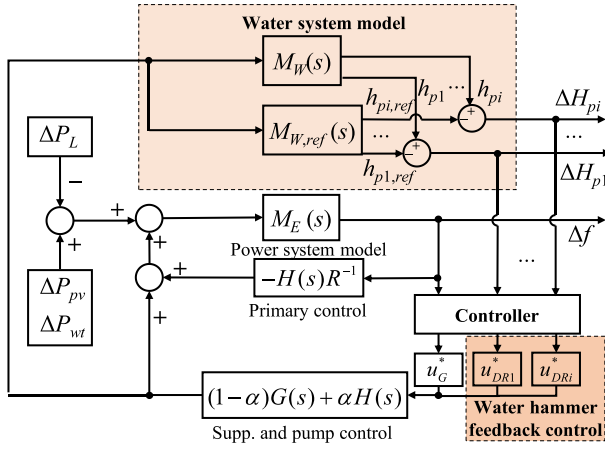


Fig. 4. IEWS-LFC model with water hammer controller for pumps.

B. IEWS-LFC Considering Water Hammer Feedback

In this subsection, dynamics of the IEWS-LFC model with water hammer feedback loop is described. Fig. 4 shows the IEWS-LFC model with water hammer controller for pumps. It contains a conventional power system controller (with control policy u_G^*) and a water hammer controller (with control policy u_{DRi}^* for each pump i). Additionally, the water system model is constructed using the dynamic response model $M_W(s)$ and steady-state model $M_{W,ref}(s)$. The performance of water hammer is evaluated by comparing the pressure h_{pi} with its steady reference value $h_{pi,ref}$. The feedback loop is composed of the water system model and water hammer feedback control, which is incorporated to suppress the water hammer effect.

The water hammer ΔH_{pi} at each pump i represents the deviation between the dynamic and steady-state pressure levels. The steady-state equation, obtained by taking the limit of the dynamic model (12), can be derived as

$$x_{ref}(t) = \bar{H}_{w,ref} x_{ref}(t-h) + \bar{B}_{w,ref} u_{DR}(t) + x_{0,ref}. \quad (13)$$

$x_{ref}(t) = [x_{ref1}(t), x_{ref2}(t), \dots, x_{refN}(t)]^T$ is the steady-state reference vector in WDS, each element contains all the pressures and volume flow rates, $x_{refi}(t) = [F_{i,ref}, f_{i,ref}, Q_{pi,ref}, F_{pi,ref}, f_{pi,ref}, h_{pi,ref}, F_{Ti,ref}, f_{Ti,ref}, h_{Ti,ref}, Q_{Ti,ref}, F_{vi,ref}, f_{vi,ref}, h_{vi,ref}]^T$. $\bar{H}_{w,ref}$ is the state transition matrix of WDS, considering the filling and draining process of the surge tank. $\bar{B}_{w,ref}$ is the control matrix of pumps. $x_{0,ref}$ is the constant term in the steady-state equation.

For the sampled-data system, the controller is combined with a sampler and a zero-order hold (ZOH). The sampler is used to sample the measurements, and the ZOH holds the control signal within a constant time interval [9]. The output of control signals to the generation and the pumps can be expressed as

$$\begin{aligned} u_G(t) &= u_G(kh), \quad kh + d_k^G \leq t < (k+1)h + d_k^G, \\ u_{DRi}(t) &= u_{DRi}(kh), \quad kh + d_{k,i}^{DR} \leq t < (k+1)h + d_{k,i}^{DR}, \\ d_m^w &\in [k_m^w h, (k_m^w + 1)h], \quad (k_m^w = 1, 2, \dots, N_m \leq N_c). \end{aligned}$$

Note that time delays $d_{k,i}^{DR}$ and d_k^G can destabilize the system or degrade the performance if not handled properly [16].

The dynamics of the sampled-data IEWS-LFC system can be formulated as

$$\dot{x}_e(t) = \bar{A}_e x_e(t) + \bar{B}_G u_G(t) + \bar{B}_{DR} u_{DR}(t) + \bar{F} \omega(t) \quad (14a)$$

$$\bar{A}_{w,0} x_w(t) + \sum_{m=1}^{N_m} \bar{A}_{w,m} x_w(t - d_m^w) + \bar{B}_w u_{DR}(t) = x_{w,0} \quad (14b)$$

$$[y_e(t), y_w(t)]^T = C[x_e(t), x_w(t)]^T \quad (14c)$$

$$x_w(t) = \varphi(t), \quad t \in [-\max d_m^w, 0). \quad (14d)$$

In (14), $x_e^T(t) = [\Delta f, \Delta P_T, \Delta P_G, \Delta P_{wt}, \Delta P_{pv}, \int \Delta f dt]$, $u_{DR}^T(t) = [u_{DR1}(t), u_{DR2}(t), \dots, u_{DRn}(t)]$, $\omega^T(t) = [\Delta P_L, \Delta P_{wind}, \Delta P_{solar}]$, $\bar{B}_G^T = [0, 0, T_g^{-1}, 0, 0, 0]$, $\bar{B}_{DR,i}^T = [(2H)^{-1}, 0, 0, 0, 0, 0]$, $\bar{B}_{DR}^T = [\bar{B}_{DR,1}^T, \bar{B}_{DR,2}^T, \dots, \bar{B}_{DR,n}^T]$,

$$\bar{A}_e = \begin{bmatrix} -D(2H)^{-1} & (2H)^{-1} & 0 & (2H)^{-1} & (2H)^{-1} & 0 \\ 0 & -T_i^{-1} & T_i^{-1} & 0 & 0 & 0 \\ -(RT_g)^{-1} & 0 & -T_g^{-1} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{wt}^{-1} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{pv}^{-1} & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\bar{F}^T = \begin{bmatrix} -(2H)^{-1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{wt}^{-1} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{pv}^{-1} & 0 \end{bmatrix}.$$

For DR i , $u_{DRi}(t) = \Delta P_{pi}$ is the output power deviation of pump i . ΔP_G is the position deviation of the speed-governor. ΔP_L , ΔP_{wind} , and ΔP_{solar} are the disturbance of the load, the wind farm, and the photovoltaic system, respectively. $\varphi(t)$ is the initial condition of WDS when $t \in [-\max d_m^w, 0)$.

III. MPC CONTROL FOR IEWS-LFC

This section focuses on the optimal control design for the demand response of IEWS. An MPC controller is developed for finding the inputs $u_G(t)$ and $u_{DR}(t)$ of IEWS-LFC.

A. Formulation

The discretized IEWS-LFC model can be described as

$$\begin{bmatrix} x_e(t_{k+1}) \\ x_w(t_k) \end{bmatrix} = A \begin{bmatrix} x_e(t_k) \\ x_w(t_k) \end{bmatrix} + \sum_{m=1}^{N_m} H_m x_w(t_k - d_m^w) + B_k u_{DR}(t_k) + \Lambda_k u_{DR}(t_{k-1}) + F_d \omega(t_k) \quad (15a)$$

$$[y_e(t_k), y_w(t_k)]^T = C[x_e(t_k), x_w(t_k)]^T \quad (15b)$$

$$x_{ref}(t_k) = \bar{H}_{w,ref} x_{ref}(t_{k-1}) + \bar{B}_{w,ref} u_{DR}(t_k) + x_{0,ref} \quad (15c)$$

$$x_w(t_k) = \varphi(t_k), \quad t \in [-\max d_m^w, 0). \quad (15d)$$

where

$$\begin{aligned} A &= \begin{bmatrix} e^{\bar{A}_e h} & 0 \\ 0 & \int_0^h \bar{A}_{w,0} ds \end{bmatrix}, \quad H_m = \int_{d_m^w - t_k^w}^h \bar{A}_{w,m} ds + \int_0^{d_m^w - t_k^w - 1} \bar{A}_{w,m-1} ds, \\ B_k &= \bar{B}_{DR} + M \bar{B}_G, \quad \Lambda_k = \bar{\Lambda}_{DR} + M \bar{\Lambda}_G, \quad M = \frac{\alpha}{1-\alpha} \bar{\Lambda}_G = \int_{h-d_k^G}^h e^{\bar{A}_e s} \bar{B}_G ds, \\ \bar{B}_G &= \int_0^{h-d_k^G} e^{\bar{A}_e s} \bar{B}_G ds, \\ \bar{\Lambda}_{DR} &= \text{diag} \{ \int_{h-d_{1,k}^{DR}}^h e^{\bar{A}_e s} \bar{B}_{DR,1} ds, \dots, \int_{h-d_{n,k}^{DR}}^h e^{\bar{A}_e s} \bar{B}_{DR,n} ds \}, \end{aligned}$$

$$\tilde{B}_{DR} = \text{diag}\left\{\int_0^{h-d_{1,k}^{DR}} \begin{bmatrix} e^{\tilde{A}_e s} \tilde{B}_{DR,1} \\ \tilde{B}_{w,1} \end{bmatrix} ds, \dots, \int_0^{h-d_{n,k}^{DR}} \begin{bmatrix} e^{\tilde{A}_e s} \tilde{B}_{DR,n} \\ \tilde{B}_{w,n} \end{bmatrix} ds\right\},$$

$$F_d = \int_0^h \begin{bmatrix} e^{\tilde{A}_e s} \tilde{F} \\ 0 \end{bmatrix} ds, x'_{w,0} = x_{w,0}h.$$

The discretization of IEWS-LFC is detailed in Appendix A.

The goal of IEWS-LFC is to find a feedback control policy $u_{DR}(t)$ to minimize the response performance index in control horizon N_c , considering the time-delays. The control policy can be expressed as

$$u_{DR}(t_k) = f(x_e(t_k), x_w(t_k) - x_{ref}(t_k), x_w(t_{k-1}), \dots, x_w(t_{k-N_m}), u_{DR}(t_{k-1}), \omega(t_k), x_{w,0}, x_{0,ref}, \text{umin } DR1, \dots, \text{umin } DRn, \text{umax } DR1, \dots, \text{umax } DRn),$$

where $\text{umin } DR, *$ and $\text{umax } DR, *$ are the control input limits of each pump.

The performance index is defined as a quadratic cost function

$$J_{N_c}(k) = \mathbb{E}\left[\sum_{m=1}^{N_c} \|x_e(t_{k+m})\|_{Q_e}^2 + \|u_{DR}(t_{k+m-1})\|_R^2 + \|x_w(t_{k+m-1}) - x_{ref}(t_{k+m-1})\|_{Q_w}^2\right]$$

$$= \mathbb{E}\left[X_e^T(k) Q_e X_e(k) + X_w^T(k) Q_w X_w(k) + U^T(k) R U(k)\right] \quad (16)$$

In (16), R is a positive definite matrix, Q_e and Q_w are the positive semi-definite matrices. $[\cdot]$ represents the mathematic expectation of $[\cdot]$. A smaller J_{N_c} implies that IEWS-LFC will reach the steady state more quickly under frequency deviation and pressure pulsation.

The selection of Q is guided by the requirements for frequency response and water hammer mitigation. These requirements are

(1) Zero steady-state frequency deviation after a load step change, i.e., $[\Delta f]^2$ [6].

(2) The time error represented by the integral of the frequency deviation should remain within ± 3 seconds, i.e., $[\int \Delta f dt]^2$ [6].

(3) Maximum pressure deviation should not exceed 30% of the rated pump outlet pressure, i.e., ΔH_{p-} [32].

Then, the weighting matrices can be defined by considering the above requirements. The MPC controller design can be written as an quadratic optimization problem

$$\min J_{N_c}(k) \quad (17a)$$

$$s.t. \begin{bmatrix} x_e(t_{k+1}) \\ x'_{w,0} \end{bmatrix} = A \begin{bmatrix} x_e(t_k) \\ x_w(t_k) \end{bmatrix}$$

$$+ \sum_{m=1}^{N_m} H_m x_w(t_{k-k_m^w}) + B_k u_{DR}(t_k) + \Lambda_k u_{DR}(t_{k-1}) + F_d \omega(t_k) \quad (17b)$$

$$x_{ref}(t_k) = \bar{H}_{w,ref} x_{ref}(t_{k-1}) + \bar{B}_{w,ref} u_{DR}(t_k) + x_{0,ref} \quad (17c)$$

$$u_{DR,i}^{\min} \leq u_{DR,i}(t_k) \leq u_{DR,i}^{\max}, \forall k = 0, \dots, N_c - 1. \quad (17d)$$

$\omega(t_k)$ is the disturbance generated by the load, wind farm, and photovoltaic system. For MPC, it is required to introduce the state values of pressure pulsations, $x_w(t_k) - x_{ref}(t_k)$, in the state equations.

Theorem 1 presents the MPC-based control policy.

Theorem 1: The control policy that minimizes the IEWS-LFC performance index $J_{N_c}(k)$ for the sampled-data IEWS model with multiple time delays can be expressed as

$$u_{DR}^*(t_k) = [I_{p \times p}, 0, \dots, 0] U_{DR}^*(k), \quad (18)$$

where p is the number of control variables. The optimal control $U_{DR}^*(k)$ for the control policy can be calculated by

$$\arg \min_{U(k)} \left[z^T(t_k) \bar{S}_k^T Q \bar{S}_k z(t_k) + U^T(k) \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + R \} U(k) + 2U^T(k) \bar{\Phi}_k^T Q \bar{S}_k z(t_k) \right] \quad (19a)$$

$$u_{DR,i}(t_k) \in \mathcal{U}_{DRi}, \forall k = 0, \dots, N_c - 1 \quad (19b)$$

$$Q = \text{diag}\{Q_e, Q_w\}. \quad (20)$$

In unconstrained finite-time MPC, the control is linear state feedback with

$$u_{DR}^*(t_k) = \Gamma(k) z^*(t_k) \quad (21)$$

$$\Gamma(k) = -[I, 0 \dots 0] \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + R \}^{-1} \bar{\Phi}_k^T Q \bar{S}_k \quad (22)$$

where $\bar{\Phi}_k = \mathbb{E}[\Phi_k]$ is the system state matrix and $\bar{S}_k = \mathbb{E}[S_k]$ is the input matrix. $z^*(t_k) = [x_e(t_k), \omega(t_k), u_{DR}^*(t_{k-1}), x_w(t_k) - x_{ref}(t_k), x_w(t_{k-1}), \dots, x_w(t_{k-N_m}), x_{w,0}, x_{0,ref}]^T$ is the feedback states at time step k . U_{DRi} is the control limit set of pump i .

These matrices can be derived from (17b) and (17c) as

$$S_k = [T_k \ T_k^2 \ \dots \ T_k^{N_c}]^T \quad (23)$$

$$\Phi_k = \begin{bmatrix} \Omega_k & \mathbf{0} & \dots & \mathbf{0} \\ T_k \Omega_k & \Omega_k & & \vdots \\ \vdots & \vdots & \ddots & \mathbf{0} \\ T_k^{N_c-1} \Omega_k & T_k^{N_c-2} \Omega_k & \dots & \Omega_k \end{bmatrix} \quad (24)$$

$$\Omega_k = \begin{bmatrix} \Omega_{e,k} \\ \Omega_{w,k} \end{bmatrix}, T_k = \text{diag}\{T_{e,k}, T_{w,k}\},$$

$$T_{e,k} = \begin{bmatrix} A_e & \Lambda_{k,e} & F_d \\ \mathbf{0} & \mathbf{0} & \Xi_k \end{bmatrix}, T_{w,k} = \begin{bmatrix} G_1 & G_2 & \varphi_1(I, \mathbf{0}) \\ \mathbf{0} & \varphi_2(I, \mathbf{0}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I \end{bmatrix},$$

$$G_1 = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \Lambda_{k,w} - \Lambda_{k,w,ref} & \bar{H}_{w,ref} \\ \Lambda_{k,w} & \mathbf{0} \end{bmatrix}, \varphi_1(I, \mathbf{0}) = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ I & -I \\ I & \mathbf{0} \end{bmatrix},$$

$$G_2 = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \dots & B \\ H_{w,1} - \bar{H}_{w,ref} & H_{w,2} & \dots & H_{w,N} \\ H_{w,1} & H_{w,2} & \dots & H_{w,N} \end{bmatrix}, \varphi_2(I, \mathbf{0}) = \begin{bmatrix} I \\ \mathbf{0} \end{bmatrix}^T$$

$$\Omega_{e,k} = \begin{bmatrix} B_{k,e} \\ \mathbf{0} \end{bmatrix}, \Omega_{w,k} = [I \ B_{k,w} - B_{w,ref} \ B_{k,w} \ \mathbf{0}]^T.$$

Ξ_k in $T_{e,k}$ is the transition matrix of the disturbance $\omega(t_k)$ such that $\omega(t_{k+1}) = \Xi_k \omega(t_k)$. Proof of Theorem 1 is given in Appendix B.

The augmented state equation contains N_c prediction steps of (17b) and (17c) ahead from the k^{th} step. Expectation calculation is applied at each step due to the uncertainty of control delays d_{ki}^{DR} and d_k^G . In practice, control delays can be estimated a priori by the statistical methods [15]. So the optimal feedback control gain $\Gamma(k)$ can be calculated offline in unconstrained condition.

Algorithm 1 MPC Controller Design

Input: The disturbance sequence $\omega(t_k)$, IEWS matrices $A, H_m, B_k, \Lambda_k, F_d$, reference matrices $\bar{H}_{w,ref}, \bar{B}_{w,ref}$, constant terms $x'_{w,0}, x_{0,ref}$, control limit U_{DRi} , and control horizon N_c

Output: Feedback control policy $u^*_{DR}(t_k)$, $k=0, 2, \dots, N_c-1$

1: Select the control weighting matrices Q_e, Q_w , and R of the MPC cost function (16).

2: Calculate the terminal constraint via (25).

3: **While** ($k \leq N_c$) **do**

4: Calculate Φ_k and $\bar{S}_k$ by (21)–(22)

5: Solve the optimization (19) to obtain the optimal control sequence $U^*_{DR}(k)$.

6: Calculate the control policy $u^*_{DR}(t_k)$ by (18)

7: $k=k+1$

8: **end while**

The design process of MPC controller is summarized in Algorithm 1.

B. Stability Analysis

Theorem 1 is only concerned with guiding the system behavior in a finite period, not about long-term stability of the system. In this subsection, the system behavior over infinite horizon is considered, and the stability of the closed-loop system will be proved using the Lyapunov method.

If IEWS-LFC is asymptotically stabilized, a control policy $u^*_{DR}(t_k)$ exist and the system state satisfying that

$$\lim_{t_k \rightarrow \infty} \mathbb{E}[\|x_e(t_k)\|_2] = 0, \quad \forall x_e(t_0),$$

$$\lim_{t_k \rightarrow \infty} \mathbb{E}[\|x_w(t_k) - x_{ref}(t_k)\|_2] = 0, \quad \forall x_w(t_0),$$

where $x_w(t) = \varphi_0(t)$, $t \in [-\max k_m^w, 0)$ denotes the initial condition of WDS. Proposition 1 and 2 are introduced based on Theorem 1.

Proposition 1: Consider the multiple-delays IEWS-LFC model. Assume that the state delays and the mean values of control delays are known. In the infinite-horizon scenario, the terminal control gain K is obtained by the following equation

$$K = [I, 0 \cdots 0] \{ \bar{\Phi}_k^T Q' \bar{\Phi}_k + \mathcal{R} \}^{-1} \bar{\Phi}_k^T Q' \bar{S}_k, \quad (25)$$

where $Q' = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & & \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & P \end{bmatrix}$, $\exists K$ with $|eig(T_k - \Omega_k K)| < 1$, such that

$$\mathbb{E}[P - (T_k - \Omega_k K)^T P (T_k - \Omega_k K)] = \mathbb{E}[Q + K^T R K]. \quad (26)$$

Proof: Along the condition (26), we have

$$\begin{aligned} z^T \mathbb{E}[P - (T_k - \Omega_k K)^T P (T_k - \Omega_k K)] z &= z^T \mathbb{E}[Q + K^T R K] z \\ &\Rightarrow \mathbb{E}[z^T P z - z^T (T_k - \Omega_k K)^T P (T_k - \Omega_k K) z] \\ &= \mathbb{E}[z^T Q z + z^T K^T R K z] \end{aligned}$$

Without loss of generality, assume $u_{DR}(t_{N_c}) = -Kz(t_{N_c})$, then we have

$$\begin{aligned} &\mathbb{E}[z(t_{N_c})^T P z(t_{N_c})] - \mathbb{E}[z^T(t_{N_c+1}) P z(t_{N_c+1})] \\ &= \mathbb{E}[z(t_{N_c})^T Q z(t_{N_c}) + u_{DR}(t_{N_c})^T R u_{DR}(t_{N_c})] \\ &\Rightarrow \mathbb{E}[z(t_{N_c})^T P z(t_{N_c})] \\ &= \mathbb{E}\left[\sum_{i=N_c}^{\infty} z(t_i)^T Q z(t_i) + u_{DR}(t_i)^T R u_{DR}(t_i)\right] \end{aligned}$$

This expression is the terminal error form in the finite horizon function $J_{N_c}(k)$, which compares with the infinite performance function $J_{\infty}(k)$.

Proposition 2: The terminal state $z(t_{k+N_c}) \subseteq \mathcal{Z}_f$, and $\mathcal{Z}_f$ is positively invariant set for system (15) with constraint (17d) under the terminal control law $u_{DR}(t_{N_c}) = -Kz(t_{N_c})$. Then, we have

$$(T_k - \Omega_k K)z(t_k) \in \mathcal{Z}_f \quad (27a)$$

$$\{-Kz(t_k)\}_i \in U_{DRi}. \quad (27b)$$

Thus, the control design of the infinite horizon can be transformed into the equivalent finite horizon optimization by replacing Q with Q' in cost function. A terminal constraint $z(t_{k+N_c}) \subseteq \mathcal{Z}_f$ is added to the optimization problem (19). It ensures that the control trajectory returns to the unconstrained MPC beyond the prediction horizon. Then, the optimal control scheme (18) can be obtained.

Consequently, stability of the MPC closed-loop control for the sampled IEWS is summarized in Theorem 2.

Theorem 2: Consider a general multiple-delay sampled IEWS (15) with $x_w(t) = \varphi_0(t)$ for $t \in [-\max k_m^w, 0)$. Assume that the state delays and the mean values of control delays are known. The IEWS is asymptotically stable with the closed-loop control scheme (18).

The proof is given in Appendix C. ■

C. Implementation Procedure

The proposed MPC controller updates the control signals in every sampling period. Its application procedure for IEWS-LFC model can be summarized as follows.

Step 1: Calculate initial flow rate and pressure, set the simulation step t_s , and the sample period h , and formulate the state-space equations.

Step 2: Find a gain K such that all eigenvalues of $T_k - \Omega_k K$ are inside the unit circle.

Step 3: Solve the terminal weighting matrix P in (26).

Step 4: Calculate the terminal constraints via (27).

Step 5: Simulate system behavior with time-step t_s .

Step 6: Calculate the optimal control policy $u^*_{DR}(t_k)$ and perform power regulation considering the control delays $d_{k,i}^{DR}$ and d_k^G .

In this process, steps 1 to 4 initialize and simulate the system dynamics. After that, at each control step t_k , the optimal control policy $u^*_{DR}(t_k)$ is determined by solving the quadratic optimization problem (17) online. System simulation continues with $u^*_{DR}(t_k)$.

TABLE I
PARAMETER VALUES IN HYDRAULIC SYSTEM

	Pipeline 1	Pipeline 2	Surge tank
Length of pipe (m)	3700	1400	-
Diameter of pipe (m)	6.8	3.8	-
Diameter of tank (m)	-	-	13
Water height in tank (m)	-	-	83
Resistance constant (s ² /m ⁵)	0.0003	0.0021	0.001

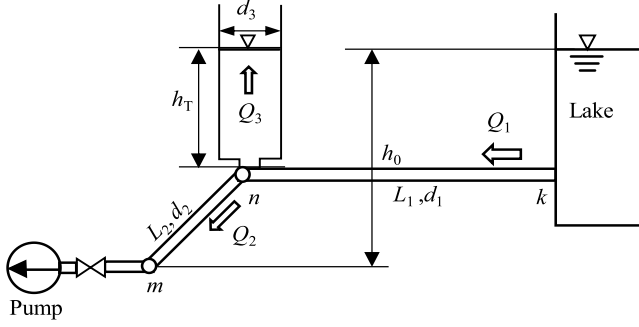


Fig. 5. Three nodes hydraulic system.

IV. CASE STUDY

The proposed MPC control of IEWS calculates the dynamic process of WDS precisely, reduces the water hammer effect, and can balance the system frequency deviation and water hammer effect. The calculation accuracy analysis was carried out on a 3-nodes hydraulic test system. The IEWS-LFC performance is verified by the test system composed of a typical WDS with 15 nodes and a single-area power system. For fair comparison, a MPC design procedure is also employed for controller design in both systems, with and without considering the water hammer effect. In addition, performance evaluation of the proposed control scheme against PI and LQR controllers is carried out in a large-scale system.

In this study, control delays vary within a certain interval and the delays are assumed to follow a uniform distribution from 0.05 to 1.2 seconds [9]. The simulation time interval and sampling period are set to 0.001s and 2s, respectively. This test was performed using MATLAB /Control System Toolbox.

A. Simulation of WDS Dynamic Process

A 3-nodes hydraulic system shown in Fig. 5 is used to evaluate the effectiveness of the proposed method. Parameters of this test system are given in Table I. Suppose that the initial flow rates Q_1 , Q_2 , and Q_3 are equal to zero. The height of water is $h_0=540$. The valve coefficient is computed from [26]

$$k_v = \frac{a\gamma}{g} A_s \sqrt{\frac{2g}{1 - (\gamma A_s)^2}}.$$

The relative area A_s is 0.087, discharge coefficient γ satisfying $\gamma=0.024t-0.00021t^2$ when $t \in [0, 40]$, $g=9.8\text{m/s}^2$, and the water hammer wave velocity a is 1250m/s.

Prediction performance between the proposed method and the finite-difference (FD) method [22] is shown in Fig. 6. h_m is the water pressure and Q_m the flow rate at the point m . Table II shows the computation time of the FD and the proposed

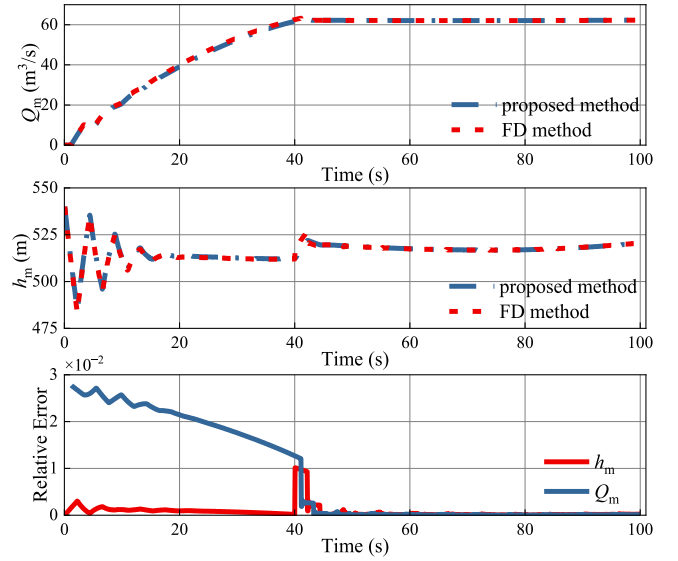


Fig. 6. Comparison of state prediction in valve m by the FD method and the proposed method.

TABLE II
COMPUTATION TIME OF FD METHOD AND PROPOSED METHOD

Methods	Proposed method	FD method
Computation Time (s)	0.052	24.6

methods. The FD method utilizes a computational grid by dividing the spatial domain into 1000 subintervals to ensure the accuracy of results. At every grid point, the solutions would be updated iteratively over time. To ensure stability, FD schemes must satisfy the Courant-Friedrichs-Lewy condition $\Delta t \leq \Delta x/a$. This requires a large number of calculations. On the other hand, the WTM considering the water hammer effect only need to utilize the flow and pressure data for calculation. It does not require to update the computational grid in every step, which significantly lowers the computation time. It can be seen from Fig. 6 and Table II that the proposed method achieves almost the same prediction results as the FD method, but takes much less calculation time (0.052 s vs. 24.6 s) to obtain the solution. The maximum relative errors $|\hat{Q}_m - Q_m|/\hat{Q}_m$ and $|\hat{h}_m - h_m|/\hat{h}_m$ between results by both methods are lower than 2.8×10^{-2} during the whole periods.

B. Advantage of Considering Water Hammer Effect

The effectiveness of IEWS control can be evaluated by comparing the water hammer and frequency performance of the proposed coordinated control method with that of the traditional LFC-DR method [10]. Parameter values of a single-area power system are given in Table III. The parameters T_g and T_t represent the equivalent time constants of the governor and turbine, respectively. R denotes the equivalent droop constant. T_{wt} and T_{pv} are the time constants of the wind turbine and photovoltaic system, respectively. H and D denote the inertia constant and the equivalent load-damping coefficient. The aggregate control delays in conventional generation and demand response (DR) are represented by dG and dDR

TABLE III
PARAMETERS OF A SINGLE-AREA POWER SYSTEM

T_g	T_l	R	$2H$
0.06 sec	0.36 sec	2.4 Hz/p.u.	0.1667 p.u.sec
D	h	T_{pv}	T_{wt}
0.0084 p.u./Hz	2 sec	1.3 sec	1.5 sec

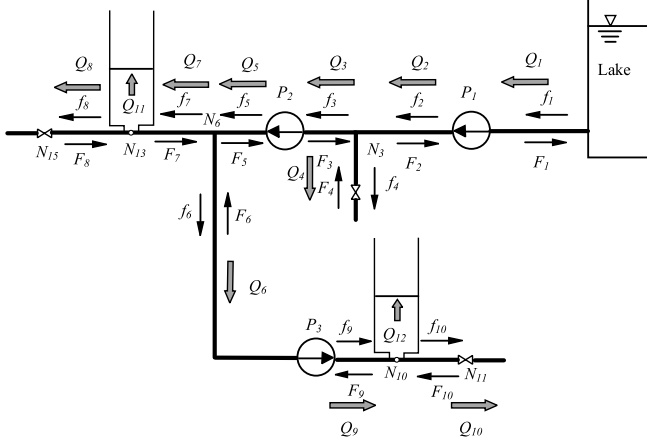


Fig. 7. Three-pump 15-node WDS wave network. The gray filled arrows denote the direction of flow. The solid one denotes the reference direction of the shock wave.

t, i , respectively. The initial pipeline flow rates and nodal pressure are determined from the steady-state hydraulic calculation. They are $Q_4=30\text{m}^3/\text{h}$, $Q_8=440\text{m}^3/\text{h}$, $Q_{10}=300\text{m}^3/\text{h}$, and $h_0=10.6\text{m}$. Fig. 7 shows the 15-node WDS modeled as a wave network [28]. There are three pumps and two surge tanks in this WDS. The initial load (p.u.) of pumps is given by $P_{p0}=[0.146, 0.141, 0.084]$. The pump efficiency is 0.8075, which is the product of converter efficiency (0.95) and motor efficiency (0.85) [4]. The DR sharing rate is set as $\alpha=0.7$. The diagonal elements of the control weighting matrix Q for Δf , $\int \Delta f dt$, and $\Delta H_{pi}=h_{pi}-h_{pi,ref}$ are all set to 1, and the control cost matrix R is $\text{diag}\{0.1, 0.1, 0.1\}$.

Consider an IEWS-LFC system with different control weighting. Set the load disturbance as 0.01 p.u., wind turbines and PV disturbance as 0 p.u. Let the aggregated delay be $d_{t,i}^{DR}=0.25\text{s}$ and $d_{t,i}^G=0.5\text{s}$. Fig. 8 illustrates the effect of frequency integral, $\int \Delta f dt$, and water hammer feedback control, ΔH_{pi} , on LFC performance. Their effects on LFC performance appear to be completely opposite. As observed in Fig. 8(a), larger penalty cost associated with $\int \Delta f dt$ exhibits higher overshoot and larger frequency fluctuation. But the system frequency is stabilized, and acts like a converged steady-state error as the water hammer penalty cost is added. It can be seen from Fig. 8(b) that their effects cancel out with simultaneous increase in costs. This demonstrates that water hammer feedback loop in the small-signal IEWS-LFC system acts as a frequency derivative feedback, capable of suppressing frequency fluctuations.

Now we evaluate the frequency deviation performance of IEWS-LFC by the proposed controller and the traditional LFC-DR control. Suppose that there are three step disturbances occurred, respectively, at (0s, 0.1p.u.) for the loads, (5s,

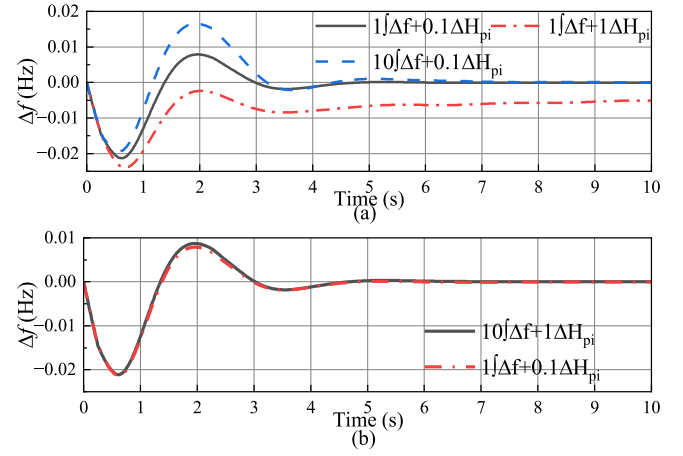


Fig. 8. Performance of frequency deviation Δf considering different state costs. (a) Various integral gain and water hammer gain. (b) Increase the frequency integral and water hammer cost by a factor of ten.

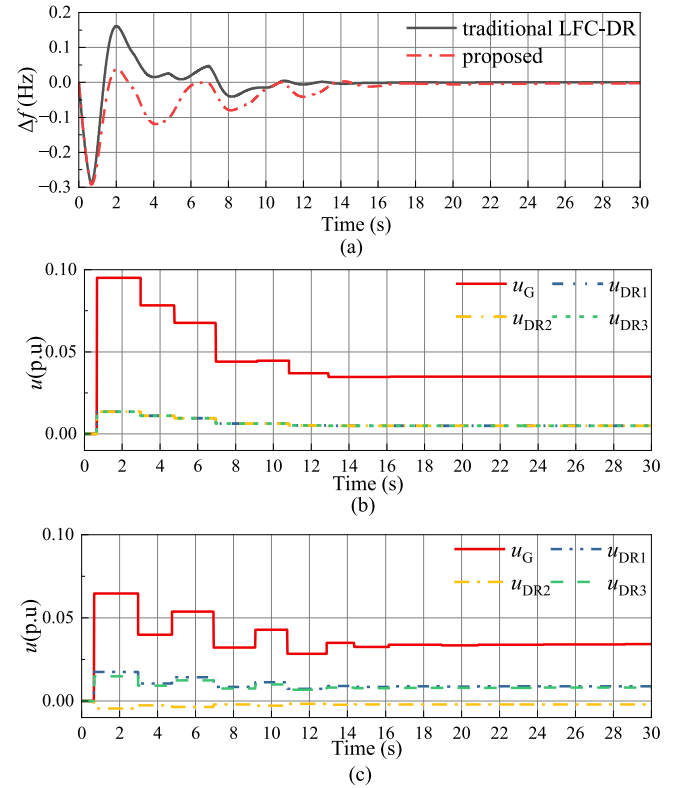


Fig. 9. Performance of frequency deviation Δf and power control inputs u_G and u_{DR} . (a) Frequency deviation performance. (b) Control inputs by traditional LFC-DR. (c) Control inputs by the proposed MPC.

0.03p.u.) for wind turbines, and (10s, 0.02 p.u.) for the PV. Fig. 9 shows the frequency deviation performance and control inputs by both methods. The traditional LFC-DR controller also used the MPC method to calculate the optimal gain, but did not consider the water hammer effect. Fig. 9(a) shows that the proposed method exhibits comparable performance with the traditional LFC-DR for a similar settling time but smaller fluctuation. It can be seen from Fig. 9(b) and 9(c) that the proposed method employs smaller control inputs than the traditional LFC-DR to achieve the same LFC performance.

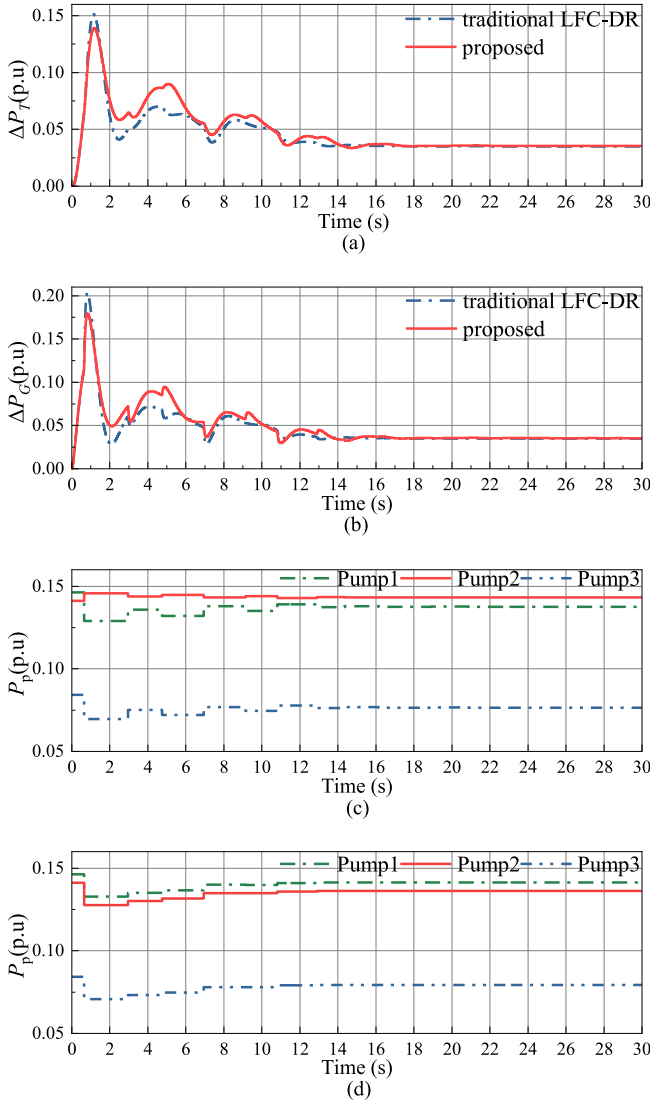


Fig. 10. Performance of ΔP_G , ΔP_T , and P_p by the proposed coordinated control and traditional LFC-DR methods. (a) Turbine output deviation ΔP_T (b) Valve position deviation of the governor ΔP_G (c) Pump power inputs by the proposed method (d) Pump power inputs by the traditional LFC-DR method.

The maximum control input by the proposed method is 0.0647 p.u., a 31.9% smaller than that by LFC-DR. Moreover, frequency fluctuation range is reduced by 27%. In addition, Fig 10 represents several time responses of the turbine output deviation ΔP_T , the governor valve position deviation ΔP_G , and the pump power inputs P_p of the IEWS. It can be seen that the proposed method in the IEWS-LFC achieves a lower peak output while maintaining similar performance comparable to the traditional LFC-DR method.

Fig. 10(c) also shows that the water hammer effect can be mitigated with different DR control inputs for different pumps. u_{DR1} , u_{DR2} and u_{DR3} denote the control inputs to pumps P_1 , P_2 , and P_3 in Fig. 7, respectively. Compared to LFC-DR, the proposed method increases the control inputs of P_1 and P_3 while reducing those for P_2 . It adjusts P_2 in the opposite direction to counteract the pressure surges caused by P_1 and P_3 . Water hammer is a transient hydraulic phenomenon

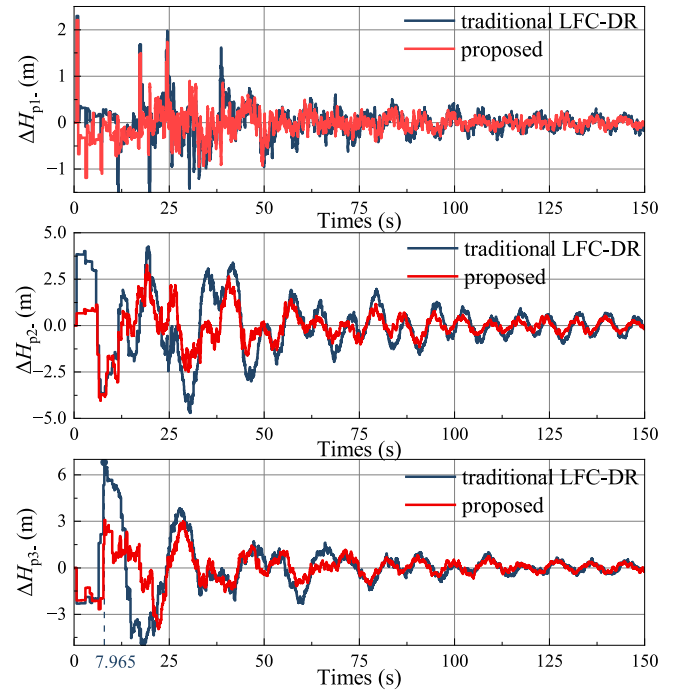


Fig. 11. Performance comparison on water hammer effect at pumps by the proposed MPC and the traditional LFC methods. Top panel: Pump p1; Mid panel: Pump p2; Bottom panel: Pump p3.

TABLE IV
RESPONSE PERFORMANCE OF TRADITIONAL LFC-DR AND PROPOSED MPC METHODS

Proposed MPC	Pump P ₁	Pump P ₂	Pump P ₃	Frequency
Settling time (s)	340	418.76	586.31	17.62
Peak pressure deviation (m)	2.20	3.26	3.10	-
Pressure fluctuation range (m)	3.40	7.33	7.05	-
system frequency nadir (Hz)	-	-	-	-0.292
Frequency deviation range (Hz)	-	-	-	0.33
Traditional LFC-DR	Pump P ₁	Pump P ₂	Pump P ₃	Frequency
Settling time (s)	382.74	469.37	999.86	16.43
Peak pressure deviation (m)	2.30	4.26	6.80	-
Pressure fluctuation range (m)	4.14	8.97	11.99	-
system frequency nadir (Hz)	-	-	-	-0.291
Frequency deviation range (Hz)	-	-	-	0.452

characterized by pressure surges and oscillations. Thus, the performance of the water hammer effect can be described by the peak pressure deviation and the time required for the system to return to steady-state. Fig. 11 shows the pressure pulsations at the outlet of the pump. The terms of ΔH_{p1-} , ΔH_{p2-} , and ΔH_{p3-} represent the pressure pulsations of P_1 , P_2 , and P_3 , respectively. It can be observed from Fig. 11 that the proposed method decreases the peak value and accelerates pressure pulsations attenuation in WDS. The cumulative effect of water pressure fluctuations at P_3 is significantly reduced, with the pressure deviation peak at 7.965 seconds for a 54.5% lower. This demonstrates that differentiated pump controls for WDS can alleviate the water hammer effect. Table IV is provided to demonstrate the advantages of the proposed method in comparison to traditional LFC-DR. It summarizes the performance by the proposed MPC and traditional LFC-DR methods. Performance measures considered are the

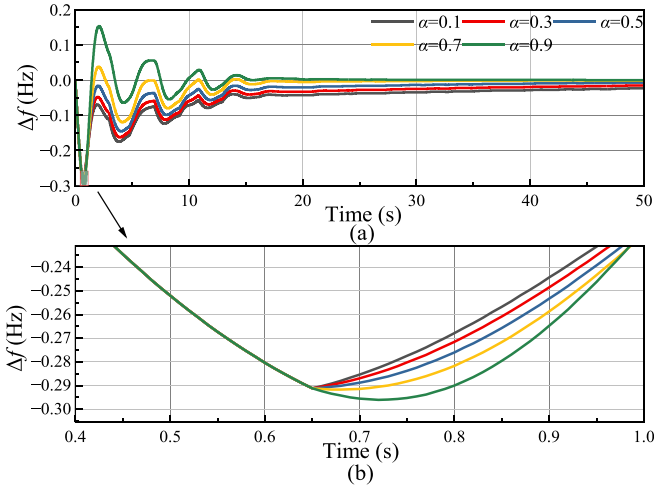


Fig. 12. Impact of the sharing rate α of the proposed IEWS-LFC model on the frequency performance.

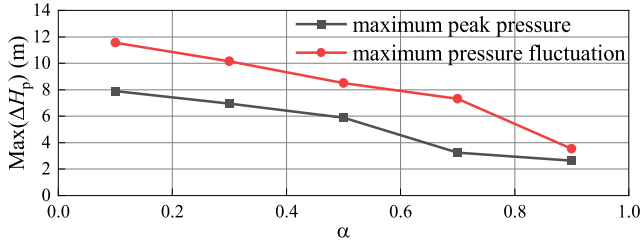


Fig. 13. Impact of the sharing rate α of the proposed IEWS-LFC model on the performance of water hammer.

peak pressure deviation ($\max\{\Delta H_{pi}\}$), the pressure fluctuation range ($\max\{\Delta H_{pi}\} - \min\{\Delta H_{pi}\}$), the system frequency nadir ($\min\{\Delta f\}$), the frequency deviation range ($\max\{\Delta f\} - \min\{\Delta f\}$), and the settling time. It can be seen from Table IV that the proposed MPC performs better than the traditional LFC-DR methods in term of water hammer mitigation and frequency fluctuation reduction. For example, peak pressure deviation and pressure fluctuation are significantly lower by the proposed MPC compared with that by traditional LFC-DR controllers. Now consider the IEWS-LFC with various DR sharing rate. The DR sharing rate $1-\alpha$ is assumed to vary in 0.1, 0.3, 0.5, 0.7 and 0.9. Frequency deviations of different sharing rates are depicted in Fig. 12. Sharing rate α indicates the proportion of the output of generation unit to the total output. A small α means that DR has great impact on the performance of LFC. It can be seen from Fig. 12(a) that small α lowers the overshoot and dampens the oscillation in the system frequency deviation.

This makes the system respond more smoothly to disturbances. But it comes at the cost of prolonged settling time. thus, it takes longer for the system to fully stabilize after a disturbance. Fig. 12(b) shows a detailed view of the frequency nadir in Fig. 12(a). Higher DR sharing rate lifts frequency nadir, which stabilizes when $1-\alpha$ exceeds 0.3. The corresponding impact on water hammer performance is shown in Fig. 13. As the DR sharing rate $1-\alpha$ increases, the regulation of pump power consumption intensifies, thereby exacerbating the water hammer phenomenon.

TABLE V
WATER HAMMER PERFORMANCE OF THE PROPOSED MPC, LQR, AND PI CONTROLLERS

Controller	Maximum peak pressure deviation (m)	Maximum pressure fluctuation range (m)	Settling time (s)
Proposed MPC	11.07	16.96	883.92
LQR	48.52	62.05	>1,000
PI	13.88	22.27	>1,000

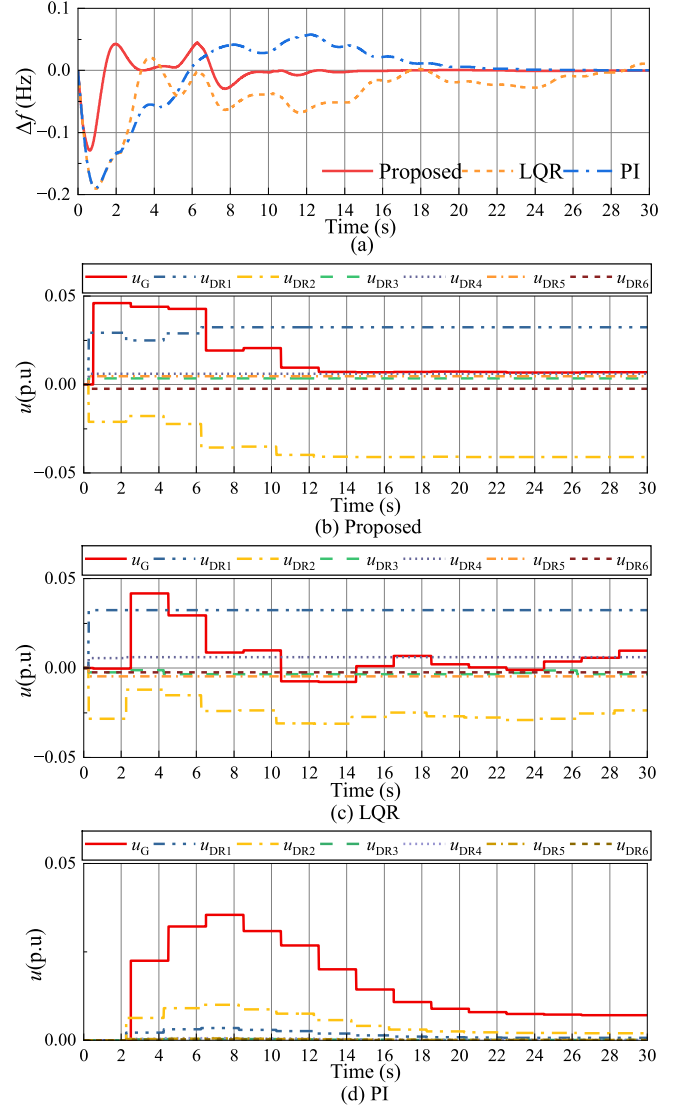


Fig. 14. Performance comparison on frequency deviation Δf and power control inputs u_G and u_{DR} by the proposed MPC, LQR, and PI controllers for a large-scale system. (a) Frequency deviation Δf . (b) Control inputs by the proposed MPC method. (c) Control inputs by the LQR controller. (d) Control inputs by the PI controller.

C. LFC Performance in Large-Scale System

A large-scale system is used to further verify the performance of the proposed control scheme. The “Richmond” water system contains 41 pipes and 6 pumps [31]. The initial electricity consumption (p.u.) of the pumps is set as $P_{p0} = [0.0807, 0.2363, 0.0085, 0.0152, 0.0117, 0.0062]$. Control input adjustment is constrained to 40% of the initial pump

load. Two surge tanks are in nodes 13 and 40. Each pump is connected to the single-area power system

LFC performance of IEWS is evaluated by three controllers: the proposed MPC controller, an LQR controller [10], and a PI controller [12] and [30]. The LQR controller considers the delay and sampling in LFC, but no constraints. The PI controller is designed using the linear matrix inequality method [30]. Although it accounts for transmission delays and the sampling process, and implements the DR based on the fair sharing principle [12], the effects of water hammer are not considered.

For the proposed MPC and LQR controllers, the weighting matrix Q for Δf and $\int \Delta f dt$ are set to 1, and the water hammer ΔH_{pi} is set to 1×10^{-4} . Three step disturbances occurred at (0s, 0.06 p.u.) for the loads, (5s, 0.03 p.u.) for the wind turbines, and (10s, 0.02 p.u.) for the PV. Coefficients for the PI controller are $K_P = 0.011$ and $K_I = 0.028$ [30].

Fig. 14 displays the frequency deviation performance Δf and control inputs u_G , u_{DR} of the IEWS by the proposed MPC, LQR, and PI controllers. Table V shows the water hammer performance. It can be seen from Fig. 14 that the proposed MPC method demonstrates better performance in frequency response deviation than other controllers. The response is faster with higher nadir. Δf decays to zero in 15 s using the proposed MPC. It takes 24 s by the PI controller and the LQR controller fails to achieve stability. The proposed MPC controller also provides better performance in mitigating water hammer, as evidenced in Table V. It also achieves a lower peak pressure deviation and a shorter settling time. These results demonstrate that the proposed MPC controller can regulate the load frequency and water hammer more efficiently than other controllers.

V. CONCLUSION

This paper has presented a controller design of an integrated electrical-water system for balancing the frequency deviation and pressure fluctuation. The computational efficiency of hydrodynamic process is obtained by eliminating unnecessary spatial discretization using the water tracking method. Based on the model predictive control, a stochastic Lyapunov equation is established to ensure the frequency stability over an infinite horizon. An analytical feedback control gain is also derived.

Simulation results have demonstrated that the proposed hydrodynamic transient modeling method is effective and the coordinated control strategy can suppress the water hammer effect. Performance of the frequency response is also improved in comparison to the traditional LFC-DR.

In future work, effort will focus on the coordinated control strategy between different generation units and the water system. Additionally, controller parameter tuning methods will be explored to improve the frequency regulation performance.

APPENDIX A

DISCRETIZATION PROCESS OF THE CONTINUOUS-TIME IEWS

In the time interval $[t_k, t_{k+1})$, the sampled-data IEWS-LFC (13) can be articulated as follow:

$$\begin{aligned} & \int_{t_k}^{t_{k+1}} \frac{d}{dt} \left(e^{-\bar{A}_e t} x_e(t) \right) dt \\ &= \int_{t_k}^{t_{k+1}} e^{-\bar{A}_e t} [\bar{B}_G u_G(t) + \bar{B}_{DR} u_{DR}(t) + \bar{F} \omega(t)] dt \\ & \sum_{m=1}^{N_m} \int_{t_k}^{t_{k+1}} \bar{A}_{w,m} x_w(t - d_m^w) dt \\ &= \int_{t_k}^{t_{k+1}} [x_{w,0} - \bar{A}_{w,0} x_w(t) - \bar{B}_w u_{DR}(t)] dt. \end{aligned}$$

The disturbance $\omega(t)$ is assumed to remain constant in sampling process. Due to the pressure wave delays dw $m \in (kw mh, (kw m+1)h]$, $kw m \geq 1$, the WDS state variables x_w are approximately unchanged during the sampling period. Then, we have

$$\begin{aligned} x_e(t_{k+1}) &= \int_{t_k}^{t_{k+1}} e^{\bar{A}_e(t_{k+1}-t)} \bar{F} dt \omega(t_k) + e^{\bar{A}_e(t_{k+1}-t_k)} x_e(t_k) \\ &+ \int_{t_k}^{t_k+d_k^G} e^{\bar{A}_e(t_{k+1}-t)} \bar{B}_G dt u_G(t_{k-1}) \\ &+ \int_{t_k+d_k^G}^{t_{k+1}} e^{\bar{A}_e(t_{k+1}-t)} \bar{B}_G dt u_G(t_k) \\ &+ e^{\bar{A}_e t_{k+1}} \text{diag} \left\{ \int_{t_k}^{t_k+d_{k,1}^{DR}} e^{-\bar{A}_e t} \bar{B}_{DR1} dt, \dots, \right. \\ &\left. \int_{t_k}^{t_k+d_{k,n}^{DR}} e^{-\bar{A}_e t} \bar{B}_{DRn} dt \right\} u_{DR}(t_{k-1}) \\ &+ e^{\bar{A}_e t_{k+1}} \text{diag} \left\{ \int_{t_k+d_{k,1}^{DR}}^{t_{k+1}} e^{-\bar{A}_e t} \bar{B}_{DR1} dt, \dots, \right. \\ &\left. \int_{t_k+d_{k,n}^{DR}}^{t_{k+1}} e^{-\bar{A}_e t} \bar{B}_{DRn} dt \right\} u_{DR}(t_k) \end{aligned} \quad (28)$$

$$\begin{aligned} & \sum_{m=1}^{N_m} \int_{t_k}^{t_{k+1}} \bar{A}_{w,m} x_w(t - d_m^w) dt = \int_{t_k}^{t_{k+1}} [x_{w,0} - \bar{A}_{w,0} x_w(t)] dt \\ & - \text{diag} \left\{ \int_{t_k}^{t_k+d_{k,1}^{DR}} \bar{B}_{w,1} dt, \dots, \int_{t_k}^{t_k+d_{k,n}^{DR}} \bar{B}_{w,n} dt \right\} u_{DR}(t_{k-1}) \\ & - \text{diag} \left\{ \int_{t_k+d_{k,1}^{DR}}^{t_{k+1}} \bar{B}_{w,1} dt, \dots, \int_{t_k+d_{k,n}^{DR}}^{t_{k+1}} \bar{B}_{w,n} dt \right\} u_{DR}(t_k). \end{aligned} \quad (29)$$

Substituting $s=t_{k+1}-t$ into (28),(29) and $v_m=t-dw$ m into (29) yield

$$\begin{aligned} x_e(t_{k+1}) &= \int_0^h e^{\bar{A}_e s} \bar{F} ds \omega(t_k) + e^{\bar{A}_e h} x_e(t_k) \\ &+ \int_{h-d_k^G}^h e^{\bar{A}_e s} \bar{B}_G dt u_G(t_{k-1}) \\ &+ \int_0^{h-d_k^G} e^{\bar{A}_e s} \bar{B}_G dt u_G(t_k) \\ &+ \text{diag} \left\{ \int_{h-d_{k,1}^{DR}}^h e^{\bar{A}_e s} \bar{B}_{DR1} dt, \dots, \right. \\ &\left. \int_{h-d_{k,n}^{DR}}^h e^{\bar{A}_e s} \bar{B}_{DRn} dt \right\} u_{DR}(t_{k-1}) \\ &+ \text{diag} \left\{ \int_0^{h-d_{k,1}^{DR}} e^{\bar{A}_e s} \bar{B}_{DR1} dt, \dots, \right. \end{aligned}$$

$$\begin{aligned}
& \int_0^{h-d_{k,n}^{DR}} e^{\bar{A}_e s} \bar{B}_{DRn} dt \Big\} u_{DR}(t_k), \\
& \sum_{m=1}^{N_m} \int_{t_k-d_m^w}^{t_k-k_m^w} \bar{A}_{w,m} dv_m x_w(t_k-k_m^w-1) \\
& + \sum_{m=1}^{N_m} \int_{t_k-k_m^w}^{t_k+1-d_m^w} \bar{A}_{w,m} dv_m x_w(t_k-k_m^w) \\
& = x_{w,0} h - \int_0^h \bar{A}_{w,0} ds x_w(t_k) - \text{diag} \left\{ \int_{h-d_{k,1}^{DR}}^h \bar{B}_{w,1} ds, \dots, \right. \\
& \left. \int_{h-d_{k,n}^{DR}}^h \bar{B}_{w,n} dt \right\} \\
& u_{DR}(t_{k-1}) - \text{diag} \left\{ \int_0^{h-d_{k,1}^{DR}} \bar{B}_{w,1} dt, \dots, \right. \\
& \left. \int_0^{h-d_{k,n}^{DR}} \bar{B}_{w,n} dt \right\} u_{DR}(t_k).
\end{aligned}$$

The discrete delay $k_{m+1}^w = k_m^w + 1$, defining $k_0^w = d_0^w = 0$, then we have

$$\begin{aligned}
\begin{bmatrix} x_e(t_{k+1}) \\ x'_{w,0} \end{bmatrix} &= A \begin{bmatrix} x_e(t_k) \\ x_w(t_k) \end{bmatrix} + \sum_{m=1}^{N_m} H_m x_w(t_k-k_m^w) + B_G u_G(t_k) \\
&+ \Lambda_G u_G(t_{k-1}) + B_{DR} u_{DR}(t_k) + \Lambda_{DR} u_{DR}(t_{k-1}) + F_d \omega(t_k)
\end{aligned}$$

At any time interval, control actions by the generator and the pumps have the following relationship

$$\frac{\alpha}{1-\alpha} u_G(t_k) = u_{DR}(t_k) \text{ and } \mu_G(t_k) = u_{DR}(t_k).$$

Then, we obtain (15a).

APPENDIX B

PROOF FOR OPTIMAL CONTROL POLICY

Along the state functions (17b) and (17c), defining $\cdot(n|k)$ as prediction of n steps ahead from time k , we have

$$\begin{aligned}
\begin{bmatrix} x_e(n+1|k) \\ x'_{w,0} \end{bmatrix} &= A \begin{bmatrix} x_e(n|k) \\ x_w(n|k) \end{bmatrix} + \sum_{m=1}^{N_m} H_m x_w(n-k_m^w|k) \quad (30a) \\
x_{ref}(n|k) &= \bar{H}_{w,ref} x_{ref}(n-1|k) + \bar{B}_{w,ref} u_{DR}(n|k) + x_{0,ref} \quad (30b)
\end{aligned}$$

Substituting $x_\Delta(n|k) = x_w(n|k) - x_{ref}(n|k)$ into (30), derive that

$$z(n+1|k) = T_k z(n|k) + \Omega_k u_{DR}(n|k) \quad (31)$$

where $z(n|k) = [x_e(n|k), \omega(n|k), u_{DR}(n-1|k), x_\Delta(n-1|k), x_w(n-1|k), \dots, x_w(n-N_m|k), x'_{w,0}, x_{0,ref}]^T$.

Consider the LFC constraints (30) in control horizon N_c , the compact form can be expressed as

$$Z(k) = S_k z(n|k) + \Phi_k U(k) \quad (32)$$

where $Z(k) = [z(1|k), \dots, z(N_c|k)]^T$, $U(k) = [u_{DR}(0|k), \dots, u_{DR}(N_c-1|k)]^T$.

Let $J_{N_c}(k) \triangleq \mathbb{E}[Z^T(k) Q Z(k) + U^T(k) \mathcal{R} U(k)]$. Using (30) to replace $Z(k)$, then we have

$$\begin{aligned}
J_{N_c}(k) &= \mathbb{E}[Z^T(k) Q Z(k) + U^T(k) \mathcal{R} U(k)] \\
&= z^T(t_k) \bar{S}_k^T Q \bar{S}_k z(t_k) + U^T(k) \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + \mathcal{R} \} U(k) \\
&\quad + 2U^T(k) \bar{\Phi}_k^T Q \bar{S}_k z(t_k)
\end{aligned}$$

This expression is an quadratic constrained optimization model with control input $U(k)$ as decision variables, aiming to minimize the cost function $J_{N_c}(k)$. Then, the optimal control scheme (18) can be calculated by solving the quadratic constrained optimal problem

$$\begin{aligned}
& \arg \min_{U(k)} \left[z^T(t_k) \bar{S}_k^T Q \bar{S}_k z(t_k) + U^T(k) \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + \mathcal{R} \} U(k) \right. \\
& \quad \left. + 2U^T(k) \bar{\Phi}_k^T Q \bar{S}_k z(t_k) \right] \\
& \text{s.t. } u_{DR,i}^{\min} \leq u_{DR,i}(t_k) \leq u_{DR,i}^{\max}, \forall k = 0, \dots, N_c - 1
\end{aligned}$$

Specially, When constraints are inactive during the control process, it degenerates into unconstrained MPC and can be derived based on the extreme value theorem, where $\nabla_{U(k)} J_{N_c}(k) = 0$.

$$\begin{aligned}
\frac{\partial J}{\partial U} \Big|_{U=U^*} &= 0 \Rightarrow 2 \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + \mathcal{R} \} U^*(k) + 2 \bar{\Phi}_k^T Q \bar{S}_k z(t_k) = 0 \\
\Rightarrow u_{DR}^*(t_k) &= -[I, 0 \dots 0] \{ \bar{\Phi}_k^T Q \bar{\Phi}_k + \mathcal{R} \}^{-1} \bar{\Phi}_k^T Q \bar{S}_k z(t_k)
\end{aligned}$$

APPENDIX C

PROOF FOR THEOREM 2

Consider the IEWS-LFC system with optimal solution $u_{DR}^*(n|k)$ at time k . Let $\omega(t)=0$, there is at least one feasible solution $u_{DR}(n|k+1)$ at time $k+1$, satisfying

$$u_{DR}(n|k+1) = u_{DR}^*(n+1|k).$$

Along the control scheme (18), the corresponding predicted state can be obtained by

$$z(n|k+1) = z^*(n+1|k).$$

Define a Lyapunov function in infinite horizon as

$$\begin{aligned}
V_{Lya}(k) &= J_\infty^*(k) = \mathbb{E} \left[\sum_{n=1}^{\infty} \|x_e^*(n|k)\|_{Q_e}^2 \right. \\
&\quad \left. + \|x_w^*(n-1|k) - x_{ref}^*(n-1|k)\|_{Q_w}^2 + \|u_{DR}^*(n-1|k)\|_R^2 \right]
\end{aligned}$$

The first forward difference of the times along the optimal control trajectory is

$$\begin{aligned}
\mathbb{E}[\Delta V_{Lya}(k)] &= \mathbb{E}[V_{Lya}(k+1) - V_{Lya}(k)] \leq \mathbb{E}[J_\infty(k+1) - J_\infty^*(k)] \\
&= \mathbb{E} \left[\sum_{n=1}^{\infty} \|x_e(n+1|k+1)\|_{Q_e}^2 - \|x_e^*(n+1|k)\|_{Q_e}^2 \right. \\
&\quad + \|x_w(n+1|k+1) - x_{ref}(n+1|k)\|_{Q_w}^2 \\
&\quad - \|x_w^*(n+1|k) - x_{ref}^*(n+1|k)\|_{Q_w}^2 + \|u_{DR}(n+1|k+1)\|_R^2 \\
&\quad \left. - \|u_{DR}^*(n+1|k)\|_R^2 \right] \\
&= \mathbb{E} \left[\sum_{n=1}^{\infty} \|x_e^*(n+1|k)\|_{Q_e}^2 - \|x_e^*(n|k)\|_{Q_e}^2 \right. \\
&\quad + \|x_w^*(n|k) - x_{ref}^*(n|k)\|_{Q_w}^2 \\
&\quad - \|x_w^*(n-1|k) - x_{ref}^*(n-1|k)\|_{Q_w}^2 + \|u_{DR}^*(n|k)\|_R^2 \\
&\quad \left. - \|u_{DR}^*(n-1|k)\|_R^2 \right]
\end{aligned}$$

The trajectory of infinite horizon follows the Riccati equation (24), thus we have

$$\mathbb{E}[\Delta V_{Lya}(k)] \leq \mathbb{E}[\|x_e^*(1|k)\|_{Q_e}^2 - \|x_w^*(0|k) - x_{ref}^*(0|k)\|_{Q_w}^2 - \|u_{DR}^*(0|k)\|_R^2]$$

Note that $Q_e > 0$, $Q_w > 0$ and $R > 0$, there must exist $\mathbb{E}[\Delta V_{Lya}(k)] < 0$.

Then, we show that the IEWS-LFC system achieves asymptotic stability if $V_{Lya}(k)$ decreases monotonically.

There is a sufficient small scalar $\delta_1 > 0$, $\delta_2 > 0$ such that

$$\mathbb{E}[\Delta V_{Lya}(k)] < -\delta_1 \mathbb{E}[x_e^T(t_k)x_e(t_k)], \mathbb{E}[\Delta V_{Lya}(k)] < -\delta_2 \mathbb{E}[(x_w(t_k) - x_{ref}(t_k))^T(x_w(t_k) - x_{ref}(t_k))].$$

Then, $\forall x_e(t_0)$, $\forall N_x \in \{1, 2, \dots\}$, such that

$$-\frac{1}{\delta_1} \sum_{k=0}^{N_x} \mathbb{E}[\Delta V_{Lya}(k)] > \sum_{k=0}^{N_x} \mathbb{E}[x_e^T(t_k)x_e(t_k)].$$

Since the $V_{Lya}(k) \geq 0$, such that

$$-\frac{1}{\delta_1} \sum_{k=0}^{N_x} \mathbb{E}[\Delta V_{Lya}(k)] = \frac{1}{\delta_1} \sum_{k=0}^{N_x} \mathbb{E}[V_{Lya}(k) - V_{Lya}(k+1)] < \frac{1}{\delta_1} \mathbb{E}[V_{Lya}(0)].$$

In a similar way, we have

$$-\frac{1}{\delta_2} \sum_{k=0}^{N_x} \mathbb{E}[\Delta V_{Lya}(k)] < \frac{1}{\delta_2} \mathbb{E}[V_{Lya}(0)].$$

Thus, we have

$$\sum_{k=0}^{N_x} \mathbb{E}[x_e^T(k)x_e(k)] < \frac{1}{\delta_1} \mathbb{E}[V_{Lya}(0)]$$

$$\sum_{k=0}^{N_x} \mathbb{E}[(x_w(k) - x_{ref}(k))^T(x_w(k) - x_{ref}(k))] < \frac{1}{\delta_2} \mathbb{E}[V_{Lya}(0)]$$

This implies that $\sum_{k=0}^{N_x} \mathbb{E}[\|x_e(t_k)\|_2]$ and $\sum_{k=0}^{N_x} \mathbb{E}[\|x_w(t_k) - x_{ref}(t_k)\|_2]$ have upper bound. Hence, the limit of $\|x_e(t_k)\|_2$ and $\|x_w(t_k) - x_{ref}(t_k)\|_2$ tend to zero, for $\forall x_e(t_0)$, $x_w(t_0)$.

APPENDIX D THE DERIVATION OF EQUATIONS

First, from (3) and (4), the shock wave equations of the i^{th} pump can be rewritten as

$$h_{pi,+}(t) = 2f_{pi,+}(t) - \frac{a}{gA_{pi,+}} Q_{pi}(t) + h_{pi,0,+} + \frac{a}{gA_{pi,+}} Q_{pi,0}, \quad (A1)$$

$$h_{pi,-}(t) = 2F_{pi,-}(t) + \frac{a}{gA_{pi,-}} Q_{pi}(t) + h_{pi,0,-} - \frac{a}{gA_{pi,-}} Q_{pi,0}. \quad (A2)$$

Substituting the pump characteristic equation

$$h_{pi,-}(t) - h_{pi,+}(t) = f(P_{pi}(t), Q_{pi}(t)) = \frac{\eta_{pi} P_{pi}(t)}{\rho g Q_{pi}(t)}$$

into (A1) and (A2), we have

$$\frac{\eta_{pi} P_{pi}(t)}{\rho g Q_{pi}(t)} - \frac{a(A_{pi,+} + A_{pi,-})}{gA_{pi,+}A_{pi,-}} Q_{pi}(t) - \Delta h_{pi,0}^* = 2(F_{pi,-}(t) - f_{pi,+}(t)) \quad (7a)$$

where $\Delta h_{pi,0}^* = h_{pi,0,-} - h_{pi,0,+} - (\frac{a}{gA_{pi,+}} + \frac{a}{gA_{pi,-}}) Q_{pi,0}$.

Adding (3) and (4) together at the inlet or outlet of the i^{th} pump, we have

$$h_{pi,+(-)}(t) = F_{pi,+(-)}(t) + f_{pi,+(-)}(t) + h_{pi,0,+(-)} \quad (7b)$$

Substituting the valve characteristic equation $Q_{vi}(t) = k_{vi}^* \sqrt{h_{vi}(t)}$ into (3) yields

$$h_{vi}(t) + k_{vi} \sqrt{h_{vi}(t)} - (h_{vi,0}^* + 2f_{vi}(t)) = 0, \quad (7c)$$

where $k_{vi} = \frac{ak_{vi}^*}{gA_{vi}}$, and

$$h_{vi,0}^* = h_{vi,0} + \frac{a}{gA_{vi}} Q_{vi,0}. \quad (7d)$$

Adding (3) and (4) together at the i^{th} valve, we have

$$F_{vi}(t) = -f_{vi}(t) + h_{vi}(t) - h_{vi,0} \quad (7e)$$

Note that, for valve, only the reflected wave of the inlet needs to be considered.

The junction of pipes is governed by the balance law of flow

$$\sum_{k \in \mathbb{J}_+} Q_k(t) = \sum_{s \in \mathbb{J}_-} Q_s(t), \quad (A3)$$

and the balance law of water head

$$h_k(t) = h_s(t), k \in \mathbb{J}_+, s \in \mathbb{J}_-. \quad (A4)$$

$\mathbb{J}_+$ denotes the junction at which the water flows in and $\mathbb{J}_-$ the junction at which the water flows out.

Subtracting (4) from (3) and rearranging, we have

$$Q_k(t) = gA_k(f_k(t) - F_k(t))/a + Q_{k,0} \quad (A5)$$

$$Q_s(t) = gA_s(f_s(t) - F_s(t))/a + Q_{s,0} \quad (A6)$$

Substituting (R5) and (R6) into (R3) yields

$$\sum_{s \in \mathbb{J}_-} \frac{gA_s(f_s(t) - F_s(t))}{a} + Q_{s,0} = \sum_{k \in \mathbb{J}_+} \frac{gA_k(f_k(t) - F_k(t))}{a} + Q_{k,0} \quad (A7)$$

Since $Q_{k,0}$ and $Q_{s,0}$ obey the flow balance law of (32), (A7) can be expressed written as

$$\sum_{k \in \mathbb{J}_+} A_k(f_k(t) - F_k(t)) = \sum_{s \in \mathbb{J}_-} A_s(f_s(t) - F_s(t)). \quad (7f)$$

Adding (3) and (4) together, we have

$$h_x(t) = f_x(t) + F_x(t) + h_{x,0}. \quad (A8)$$

Substituting (A8) into (A4), we have

$$f_k(t) + F_k(t) + h_{k,0} = f_s(t) + F_s(t) + h_{s,0}, k \in \mathbb{J}_+, s \in \mathbb{J}_-. \quad (A9)$$

Since $h_{k,0}$ and $h_{s,0}$ obey the water pressure balance law of (A4), (A9) can be written as

$$f_k(t) + F_k(t) = f_s(t) + F_s(t), k \in \mathbb{J}_+, s \in \mathbb{J}_-. \quad (7g)$$

The dynamic model at the surge tank with a T-junction, which is a special case of the pipe junction model, can be written as

$$Q_{Ti,+}(t) - Q_{Ti,-}(t) = Q_{Ti}(t). \quad (A10)$$

Substituting (A10) into the characteristic equation of surge tank

$$h_{Ti}(t + \Delta t) - h_{Ti}(t) = Q_{Ti}(t) \times \Delta t / A_{Ti},$$

we have

$$h_{Ti}(t + \Delta t) - h_{Ti}(t) = (Q_{Ti,+}(t) - Q_{Ti,-}(t)) * \Delta t / A_{Ti}. \quad (7h)$$

It follows from (A5)-(A6) and (A8) that the pressure wave at the tank is governed by

$$F_{Ti}(t) = -f_{Ti}(t) + h_{Ti}(t) - h_{Ti,0}, \quad (7i)$$

$$Q_{Ti,+(-)}(t) = \frac{gA_{Ti,+(-)}}{a} (f_{Ti,+(-)}(t) - F_{Ti,+(-)}(t)) + Q_{Ti,0,+(-)}. \quad (7j)$$

APPENDIX E

THE STRUCTURE OF THE MATRIX $\overline{A}_w$ And $\overline{B}_w$

i) The state transition matrix $\overline{A}_w$

$\overline{A}_w$ is constructed based on equation (7b), (7d)–(7j), and (8) to (11). It is a square matrix of $n_w \times n_w$, where n_w is number of state variables. The equations associated with the construction of the matrix $\overline{A}_w$ can be divided into three groups: flow balance equation (7f), pressure balance equation (7g), and component characterization equations (7b), (7d)–(7e), (7h)–(7j) and (8)–(11). The specific structure of $\overline{A}_w$ is represented as bellow:

$$\overline{A}_{w,0} = \begin{bmatrix} 2\mathbf{I}_{p \times p} & 0 & 0 & -2\mathbf{I}_{p \times p} & -\mathbf{A}_{p1} & 0 & 0 \\ \mathbf{I}_{p \times p} & 0 & \mathbf{I}_{p \times p} & 0 & 0 & \mathbf{I}_{p \times p} & 0 \\ 0 & \mathbf{I}_{p \times p} & 0 & \mathbf{I}_{p \times p} & 0 & 0 & \mathbf{I}_{p \times p} \\ \mathbf{C}_{ot}\mathbf{B}_{11} & -\mathbf{C}_{in}\mathbf{B}_{11} & -\mathbf{C}_{ot}\mathbf{B}_{11} & \mathbf{C}_{in}\mathbf{B}_{11} & 0 & 0 & 0 \\ \mathbf{D}_{1,+} & \mathbf{D}_{1,-} & \mathbf{D}_{1,+} & \mathbf{D}_{1,-} & 0 & 0 & 0 \\ 0 & \mathbf{I}_{l \times l} & 0 & 0 & \mathbf{A}_{11} & 0 & 0 \\ 0 & 0 & \mathbf{I}_{l \times l} & 0 & -\mathbf{A}_{11} & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$\overline{A}_{w,m} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\mathbf{I}_{l \times l}(t, t) & 0 & 0 & 0 & \mathbf{A}_{11}(t, t) & 0 & 0 \\ 0 & 0 & 0 & -\mathbf{I}_{l \times l}(t, t) & -\mathbf{A}_{11}(t, t) & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

The first row of the matrix $\overline{A}_{w,0}$ is obtained by equation (10) of pump output. The second and the third rows of $\overline{A}_{w,0}$ are obtained by equation (7b) about pump pressure. The fourth row and the fifth row are obtained by the flow balance equation (7f) and pressure balance equation (7g), respectively. Then, other rows are obtained by the component characterization equations (7d)–(7e), (7h)–(7j), (8)–(9), and (11). $\overline{A}_{w,m}$ is the state transition matrices with time-delay. It can be obtained by the pipe transmission equations (8)–(9) and the time-delay equation of surge tank (7h).

ii) The input matrix $\overline{B}_w$

$\overline{B}_w$ is constructed from the pump equation (10). It is a matrix of $n_w \times n_p$, where n_p is number of pumps. It is obtained by

combining the output matrices in each pump equation, which can be articulated as

$$\overline{B}_w = \begin{bmatrix} \mathbf{B}_{pw} \\ 0 \end{bmatrix},$$

$$\text{where } \mathbf{B}_{pw} = \text{diag}\left\{\frac{\eta_{p1}}{\rho g Q_{pi,0}}, \dots, \frac{\eta_{pi}}{\rho g Q_{pi,0}}\right\}.$$

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